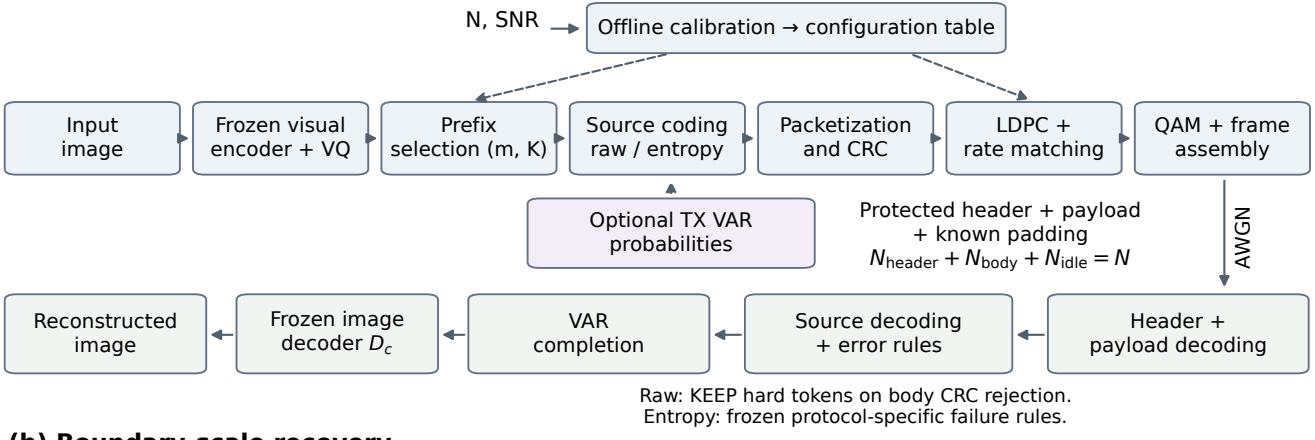
(a) Transmission and reconstruction
N, SNR
Offline calibration → configuration table
Input image
Frozen visual encoder + VQ
Prefix selection (m, K)
Source coding raw / entropy
Packetization and CRC
LDPC + rate matching
QAM + frame assembly
Optional TX VAR probabilities
Protected header + payload + known padding
$N_{\text{header}} + N_{\text{body}} + N_{\text{idle}} = N$
AWGN
Reconstructed image
Frozen image decoder D_c
VAR completion
Source decoding + error rules
Header + payload decoding
Raw: KEEP hard tokens on body CRC rejection.
Entropy: frozen protocol-specific failure rules.
(b) Boundary-scale recovery
Received coarse scale prefix
Predict boundary scale
Overwrite received boundary positions
Update accumulated latent
Predict finer scales
Received tokens condition finer scales; other positions in the same boundary scale are not recomputed.